

SOH CAH TOA

$\sin\theta = \frac{opp}{hyp}$

$90 + (90-\theta) + \alpha = 180$

$\alpha = \theta$



$g = -g\sin\theta\,\hat{e}_z + g\cos\theta\,\hat{e}_x$

Q1]



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$

Effect of gravity ✗

pressure ?



Mass: $\frac{\partial v_z}{\partial z} = 0$

Assumptions

+ Newtonian fluid

Stokes: $\mu \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \rho g_z = 0$

$\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = -\rho g_z \frac{(r)}{\mu}$

$r\frac{\partial v_z}{\partial r} = \frac{-\rho g_z}{\mu}\frac{r^2}{2} + \frac{C_1}{r}$

$v_z = -\frac{\rho g_z}{2\mu}\left(\frac{r^2}{2}\right) + C_1 \ln r + C_2$

@ $r = R$, $v_z = 0$

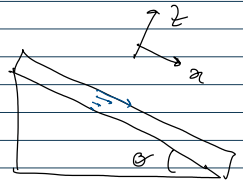
$C_2 = \left(+\frac{\rho g_z R^2}{4\mu} - C_1 \ln R\right)$

@ $r = d+R$, $\tau_{rz} = 0$; $\tau_{rz} = \mu\left(\frac{\partial v_r}{\partial z}^{0} + \frac{\partial v_z}{\partial r}\right) = 0$

$C_1 = \left(\frac{\rho g_z}{2\mu}(d+R)\right)(d+R)$

(i) $v_z\,(@\,r = R)$

Q2]

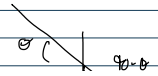


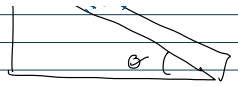
(a) $\nabla \cdot \underline{v} = 0$

$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}^{0} + \frac{\partial v_z}{\partial z}^{0} = 0$

$v = \begin{pmatrix} v_x \\ 0 \\ v_z \end{pmatrix}$ $p = ?$

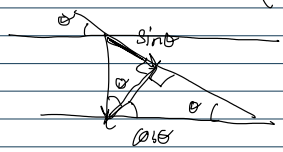
$g = \begin{pmatrix} g\sin\theta \\ 0 \\ -g\cos\theta \end{pmatrix}$

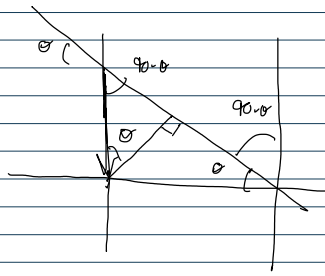




$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$g = \begin{pmatrix} g\sin\theta \\ 0 \\ -g\cos\theta \end{pmatrix}$$





(b)

$$0 = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x$$

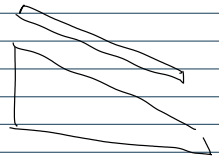
$$0 = -\frac{\partial p}{\partial y} 0$$

$$0 = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2}{\partial z^2}\right) + \rho g_z$$

(c) @ $z = 0$, $v_x = 0$

@ $z = H$, $\tau_{zx} = 0$; $\tau_{zz} =$

(d)



@ $z = H$, $v_x = 0$